

GREAT WALL STATION, Antarctica

WMO: 890580

Lat: 62.217S

Lon: 58.967W

Elev: 33

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.1	5.7	-2.3	4.9	2.5	1.6	6.0	6.8	60.4	22.0	55.6	21.9	17.6	110	1.155

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.2	40.2	38.0	39.2	37.3	38.4	36.7	38.8	39.4	37.9	38.5	37.3	37.9	17.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
38.2	33.9	38.8	37.3	32.8	38.1	36.6	31.8	37.6	14.8	39.7	14.5	38.8	14.2	38.0	43.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 57.4	48.0	40.5	DB	0.1	44.0	4.9	2.8	-3.5	46.0	-6.4	47.7	-9.2	49.3	-12.8	51.3	(4)
(5)			WB	0.0	40.1	5.0	1.1	-3.6	40.9	-6.5	41.6	-9.3	42.2	-12.9	43.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	35.1	35.3	33.0	29.4	27.0	22.9	21.6	22.7	24.3	27.2	30.4	32.8	
	DBStd	6.90	1.87	2.50	3.62	5.43	5.29	7.20	7.43	6.73	6.24	4.64	2.81	2.40	
	HDD50	7870	463	413	526	619	714	813	880	845	770	706	587	534	
	HDD65	13345	928	833	991	1069	1179	1263	1345	1310	1220	1171	1037	999	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	18.5	15.7	16.7	17.3	18.8	18.3	19.2	18.7	21.2	20.4	21.0	17.5	17.3	
Precipitation	PrecAvg	26.9	2.3	2.6	2.8	2.5	2.4	2.1	2.1	2.4	2.3	2.2	1.8	1.7	
	PrecMax	32.3	5.0	3.8	4.8	5.3	4.1	5.3	5.5	6.3	4.1	4.6	3.2	3.1	
	PrecMin	18.5	0.8	0.5	0.4	1.3	1.1	0.2	0.5	1.0	1.1	0.9	0.5	0.6	
	PrecStd	4.1	1.0	0.7	1.0	0.9	0.8	1.0	1.0	1.2	0.8	0.9	0.6	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	42.1	41.0	38.6	36.3	34.8	33.5	33.6	34.6	35.9	38.0	41.4	
		MCWB	38.7	39.4	39.2	37.4	35.0	33.8	32.5	32.5	33.3	34.3	35.6	37.3	
	2%	DB	39.6	40.2	39.3	37.3	35.2	33.6	32.6	32.4	33.1	34.0	35.8	38.7	
		MCWB	37.2	38.2	38.0	36.3	34.2	32.8	31.8	31.6	32.3	32.8	34.2	35.8	
	5%	DB	38.7	39.1	38.4	36.3	34.4	32.7	31.8	31.9	32.3	33.2	34.8	37.4	
		MCWB	36.5	37.5	37.3	35.5	33.6	32.1	31.2	31.2	31.7	32.2	33.5	35.0	
	10%	DB	37.8	38.3	37.6	35.3	33.6	31.7	30.9	31.2	31.7	32.5	34.0	36.3	
		MCWB	36.0	36.9	36.6	34.5	33.0	31.2	30.3	30.6	31.2	31.7	32.9	34.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	40.4	39.8	37.8	35.6	34.4	33.1	32.9	33.8	34.7	36.0	38.2
			MCDB	40.3	41.3	40.7	38.1	35.8	34.6	33.2	33.0	34.1	35.4	37.2	39.8
2%		WB	38.1	39.0	38.5	36.6	34.6	33.2	32.2	32.0	32.4	33.3	34.8	36.7	
		MCDB	38.8	39.6	39.0	37.0	34.9	33.4	32.4	32.1	32.6	33.7	35.5	37.9	
5%		WB	37.3	38.1	37.7	35.7	33.9	32.3	31.3	31.3	31.9	32.6	34.0	35.7	
		MCDB	38.0	38.6	38.1	36.2	34.3	32.5	31.6	31.6	32.1	33.0	34.5	36.7	
10%		WB	36.6	37.4	36.9	34.8	33.1	31.4	30.4	30.7	31.3	32.0	33.2	34.8	
		MCDB	37.4	38.0	37.4	35.3	33.5	31.7	30.9	31.1	31.6	32.4	33.8	35.8	
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.2	4.5	4.6	5.3	6.2	7.5	6.7	6.6	5.1	4.4	4.5	
		MCDBR	5.4	5.3	4.6	4.3	4.4	4.9	6.0	5.4	5.5	4.7	5.2	6.7	
		MCWBR	4.2	4.2	4.2	4.4	4.8	5.1	6.3	5.6	5.4	4.5	4.3	4.9	
	5% WB	MCDBR	4.8	4.7	4.2	4.1	4.3	4.9	6.3	5.2	5.4	4.5	4.4	5.7	
		MCWBR	4.2	4.2	4.2	4.4	4.9	5.2	6.6	5.5	5.5	4.5	4.1	4.7	
Clear-Sky Solar Irradiance	taub	0.309	0.308	0.297	0.280	0.225	0.366	0.211	0.269	0.288	0.310	0.304	0.309		
	taud	2.494	2.503	2.503	2.373	2.099	2.820	2.048	2.239	2.370	2.402	2.458	2.467		
	Ebn at Noon	307	291	267	214	140	95	139	209	263	286	306	310		
	Edh at Noon	31	28	23	17	10	6	10	18	25	30	32	33		
All-Sky Solar Radiation	RadAvg	1595	1151	619	271	78	33	56	201	530	984	1508	1807		
	RadStd	145	88	51	22	7	1	3	17	42	65	91	144		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	-0.47	-0.76	N/A	N/A	N/A	N/A	

Nomenclature: See separate page